

Tutorial 2

Econometric Theory II

Francesco Sbrana

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Selection Bias is constant over time

- Many reasons NOT to believe in PTA

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- 2 Selection Factors may depend on temporal shocks
- 3 PTA depends on the functional form assumed by the outcome. PTA might be violated depending on the form

What we can do more to believe PTA?

1) **Pre-trend testing**

we already saw it, let's see an example to refresh our memory

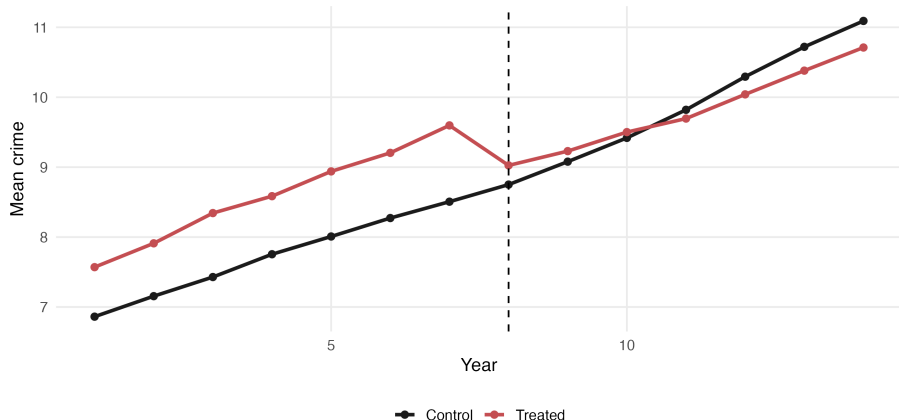
Data

- 10,000 units observed over 14 years (140,000 rows)
- 30% treated
- $\mathbf{i}$ = City, $\mathbf{Y}$ = Crime
- Treatment starts in year 8
- Two discrete covariates only: region, risk
- **Rules:**
 - Unconditional parallel trends fails
 - Conditional parallel trends holds given (region, risk)

Unconditional Outcome Plot

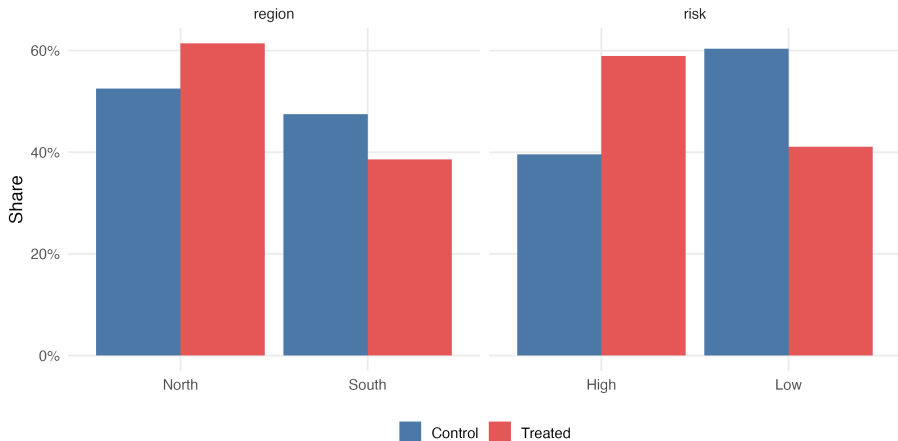
Unconditional mean crime by treatment status

Raw trends are not parallel because treatment targets high-risk covariate cells



Covariate Plot

Covariate balance before adjustment



Balance Test

=====

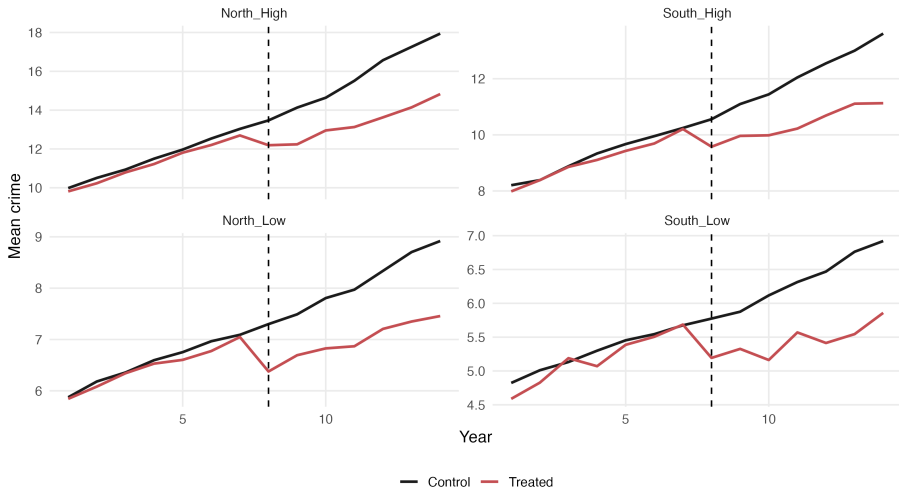
PART B SUMMARY – COVARIATE BALANCE

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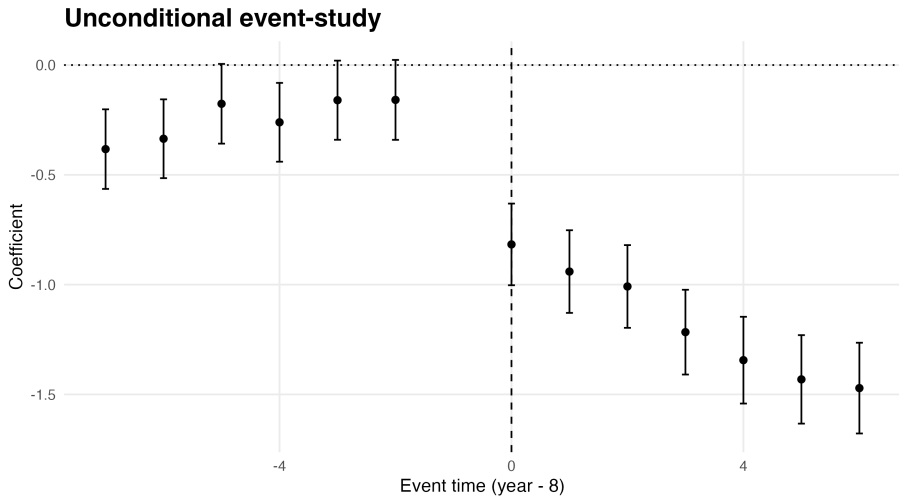
```
> print_table(balance_tests)
covariate statistic  p_value
region      66.98 2.748e-16
risk       315.40 1.459e-70
> cat("Interpretation: small p-values indicate treated and control groups differ in covariate composition before adjustment.\n")
Interpretation: small p-values indicate treated and control groups differ in covariate composition before adjustment.
>
```

Conditional Outcome Plot

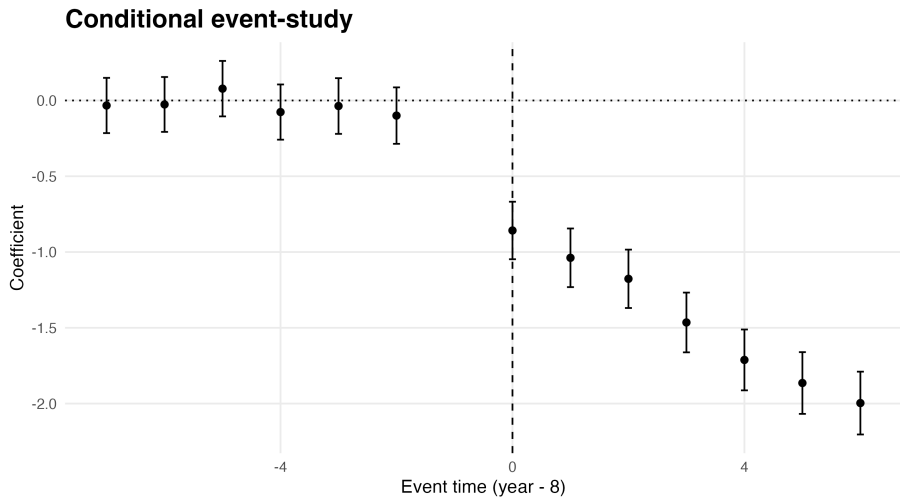
Within-cell trends by treatment status



Unconditional Event Study



Unconditional Event Study



Look at the Pre-Trends

PART C SUMMARY – CONDITIONAL PARALLEL TRENDS

```
> cat("Key pre-treatment coefficients from conditional event study:\n")
Key pre-treatment coefficients from conditional event study:
> print_table(cond_prel, c("rel_year", "estimate", "std.error", "conf.low", "conf.high"))
rel_year estimate std.error conf.low conf.high
-7 -0.03307 0.09298 -0.2153 0.14919
-6 -0.02619 0.09238 -0.2073 0.15490
-5 0.07801 0.09336 -0.1050 0.26102
-4 -0.07650 0.09293 -0.2587 0.10567
-3 -0.03655 0.09377 -0.2204 0.14726
-2 -0.09979 0.09501 -0.2860 0.08645
> cat("Interpretation: after conditioning on covariate cells, pre-treatment coefficients should move closer to zero if conditional PT is plausible.\n")
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```

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Limitations:

- Parallel pre-trends do not necessarily imply parallel counterfactual post-treatment trends
- We can mistakenly conclude that pre-trends are parallel:
 - **Low power** → fail to detect pre-trend violations even though those should matter
 - **Pre-testing bias** → selection bias (*pre-test bias*) that exaggerate post effects

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To make pre-trend testing more credible:

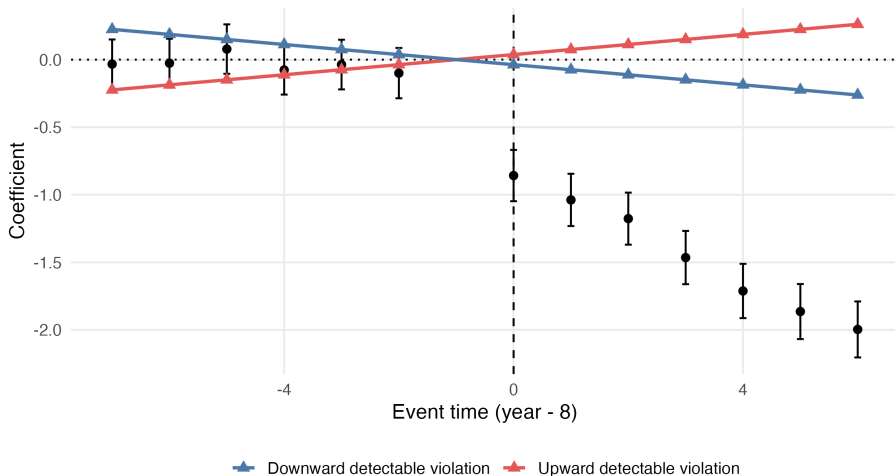
- **Roth (2022)** → *What is the smallest linear violation of parallel trends that I can detect with a test with 80% power?*
- **Rambachan and Roth (2023)** → *How large would violations need to be to overturn our conclusions?*

How it works (simple intuition):

- Compute pre-trends
- Using these estimates, compute the smallest pre-trend violation you are able to detect with power 80% (call it δ)
- Compute $\frac{\sum_{t=1}^T \delta * t}{T}$
- That is the part of the coefficient magnitude which can be due to bias

Roth (2022): detectable linear violations

Both upward- and downward-sloping violations with 80% power



Main question:

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- **Relative magnitude** → Post-treatment violations cannot be more than M times larger than pre-treatment deviations

$$|\text{post-treatment violation}| \leq M \times \max |\text{pre-trend}|$$

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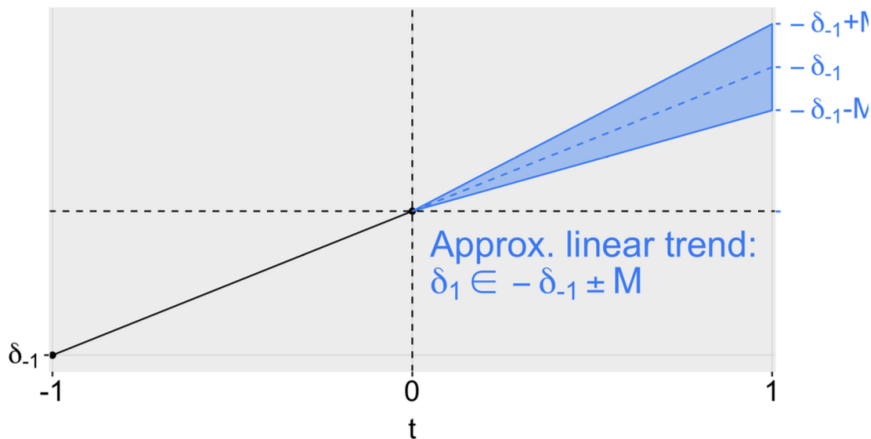
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- **Smoothness Restriction** → Imposes bounds on trend changes

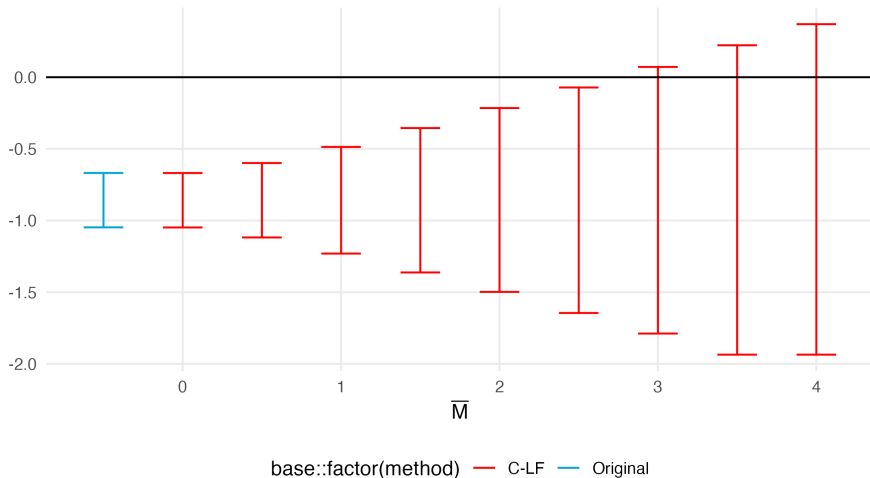
$$|\Delta^2 \text{trend}| \leq M$$

$$\delta_1 \in [-\delta_1 - M, -\delta_1 + M]$$



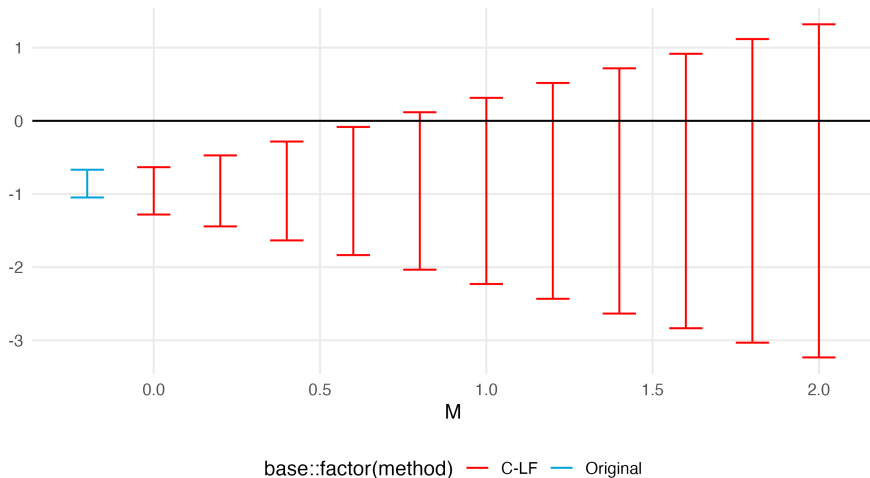
HonestDiD sensitivity plot (relative magnitudes)

Conditional event-study specification



HonestDiD sensitivity plot (smoothness restriction)

Conditional event-study specification



Covariates (again)

Why to include covariates in DiD?

- PTA may fail unconditionally, but hold conditional on covariates

Conditional PTA: *within groups defined by the same X , treated and control units would have...*

How to insert covariates?

If we just put covariates in the TWFE regression, we are implicitly assuming the functional form and homogenous effects. It might not be the case.

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We need an estimator that adjust for X:

- **IPW** (Inverse Probability Weighting, Abadie 2005)
- **OR** (Outcome Regression, Heckman et al. 1997)
- **DR** (Doubly Robust, Sant'Anna and Zaho 2021)

IPW

IDEA: reweight the control group so that they resemble the pre-treatment covariate distribution of the treated group

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$$\theta_{IPW} = E[Y_{t=1} - Y_{t=0} \mid W = 1] - E_{\omega}[Y_{t=1} - Y_{t=0} \mid W = 0]$$

where:

$$E_{\omega}[Y_{t=1} - Y_{t=0} \mid W = 0] = E\left[\frac{e(X)}{1 - e(X)} (Y_{t=1} - Y_{t=0}) \mid W = 0\right]$$

$e(X) = P(W = 1 \mid X) \rightarrow$ propensity score

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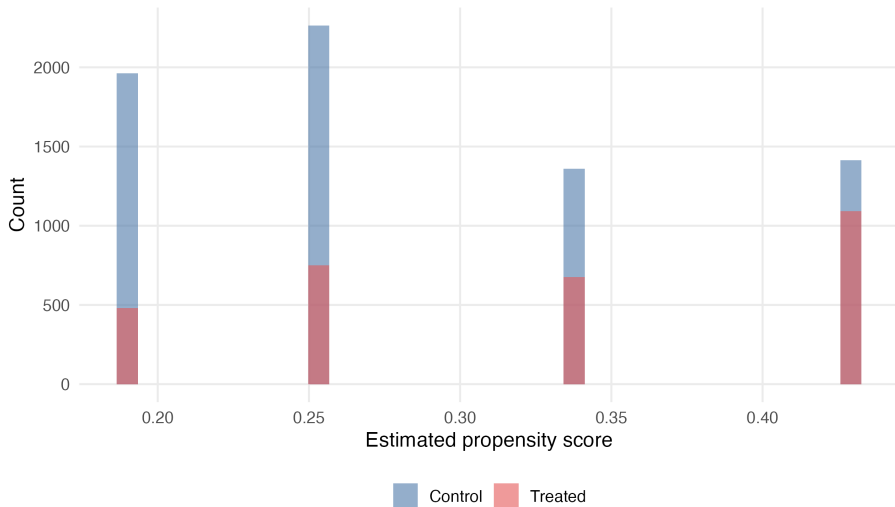
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Assumptions:

- Conditional PTA
- Common Support $\rightarrow 0 < Pr(W = 1|X) < 1$
- Correct propensity score specification

Estimated propensity score overlap



OR

IDEA: estimate potential outcome $Y(0)$ for the treated group using regression among controls on covariates.

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$$\theta_{OR} = E[Y_{t=1} - Y_{t=0} \mid W = 1] - E[\hat{m}(X, t = 1) - \hat{m}(X, t = 0) \mid W = 1]$$

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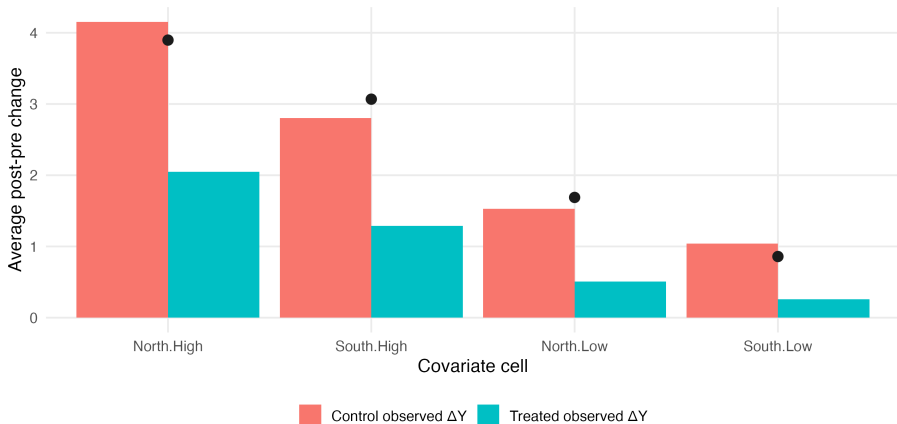
$$\hat{m}(X, t) = E[Y_t \mid X, W = 0]$$

- **Steps:**

1. Regress ΔY (or Y_t separately) on baseline X among **control units only**.
2. Get fitted values using X from **treated units** $\Rightarrow \widehat{\Delta Y}(0)$ for $W = 1$
3. $ATT = \text{actual } \Delta Y \text{ for treated} - \text{imputed } \widehat{\Delta Y}(0) \text{ for treated}$

Outcome-regression intermediate step

Bars: observed mean ΔY by cell; black dots: predicted untreated $\Delta Y(0)$ for treated cells



IDEA: use both methods! If one is not correct but the other is, you get the correct estimate

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$$\begin{aligned} \theta_{DR} = & \left(E[Y_{t=1} - Y_{t=0} \mid W = 1] \right. \\ & \left. - E[\hat{m}(X, t = 1) - \hat{m}(X, t = 0) \mid W = 1] \right) \\ & - \\ & \left(E\left[\frac{e(X)}{1 - e(X)} (Y_{t=1} - Y_{t=0}) \mid W = 0 \right] \right. \\ & \left. - E\left[\frac{e(X)}{1 - e(X)} (\hat{m}(X, t = 1) - \hat{m}(X, t = 0)) \mid W = 0 \right] \right) \end{aligned}$$

Comparison

Comparing ATT estimates

TWFE unconditional vs Abadie/IPW vs Outcome Regression vs Doubly Robust

